

Supporting Information for

Unifying the Gutenberg-Richter Law with Probabilistic Catalog Completeness

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Introduction

This Supporting Information provides detailed descriptions of the data, methods, and results that support our proposed probabilistic framework for modeling catalog incompleteness using four augmented Gutenberg-Richter models. It includes a description of the fitting algorithm used to ensure numerical stability and robustness, evaluations of model performance on both synthetic and empirical earthquake catalogs, and seismological interpretations of key parameter estimates. The figures include probability density functions (PDFs) of the estimated parameters across six study regions, modeled and observed cumulative frequency-magnitude distributions, and residual diagnostics. The tables summarize parameter estimates and performance metrics for both synthetic and empirical datasets.

Text S1. Fitting algorithm for the augmented Gutenberg-Richter models

To enhance both computational efficiency and numerical robustness, particularly to mitigate the impact of tail-region residual perturbations (Figure S9), we employ a weighted residual nonlinear least squares fitting method. The weighting scheme is based on the square root of the normalized complementary cumulative distribution function, i.e., $\sqrt{\log_{10} \text{CCDF}(m)}$, which assigns greater weight to the more stable, high-frequency regions without allowing extreme values to dominate the fit. This strategy strikes a balance between emphasizing reliable regions of the distribution and accounting for the heteroscedastic nature of frequency-magnitude distribution, where variance decreases as frequency increases. To further avoid local minima, five parallel optimizations are performed using randomly perturbed initial parameter sets, and the solution with the lowest weighted residual is retained.

Text S2. Performance on simulated catalog

Tables S1(a) to S4(a) show, overall, as the sample size parameter k increases, the fitted parameters $\theta^{\text{pred}} = \{a^{\text{pred}}, b^{\text{pred}}, m^{\text{pred}}, \sigma_c^{\text{pred}}, \zeta^{\text{pred}}\}$ progressively converge toward their predefined true values. Simultaneously, the uncertainty associated with the parameter estimates, quantified by the standard deviations from 200 independently sampled datasets, tends to decrease with increasing k .

To provide further insight, the statistics are broken down into two categories for the simulated catalog: (1) correct model specification, cases where each model is fitted to data generated by itself, corresponding to the light red-shaded rows in Tables S1(a) to S4(a), and (2) model mis-specification, cases where a model is fitted to data generated by a different model, corresponding to the light blue-shaded rows in Tables S1(a) to S4(a). In the cases of correct model specification, all models exhibit favorable performances (1st or 2nd place).

In the cases of model mis-specification, all models exhibit a decrease in performance compared to correct model specification. Among these models, GR_{BS} and GR_{CO} generally yield weaker results, often ranked as poor (light blue) or worst (dark blue) across all evaluation metrics. GR_{SS} exhibits intermediate performance. GR_{AE} is most often ranked first or second, respectively, with no occurrences of being ranked as worst. These results indicate the superior adaptability of GR_{AE} , especially under conditions of model mis-specification. This can be attributed to the greater flexibility and smoothness of GR_{AE} ; notably, GR_{CO} is a special case of GR_{AE} with the shape parameter $\zeta = 0$, limiting its adaptability.

Text S3. Performance on empirical catalog

Table 1 shows the fitted parameters of the augmented Gutenberg-Richter models applied to empirical earthquake catalogs from six study regions, including four regions in China (Beijing-Tianjin-Hebei, the southeastern coast, Sichuan-Yunnan, and northern Xinjiang), as well as California and New Zealand. The values in Table 1 are the means and standard deviations derived from 200 bootstrap iterations. Each bootstrap iteration involves resampling the empirical catalog with replacement to generate a dataset of equal size, resulting in 200 distinct catalog realizations.

Based on the standard deviations of m_c^{pred} , a^{pred} , b^{pred} , and σ_c^{pred} (Table 1; Figures S1 to S4), GR_{AE}, GR_{SS} and GR_{CO} generally exhibit small parameter estimation uncertainties. The GR_{BS} model consistently shows the highest uncertainty, with standard deviations typically an order of magnitude larger than those of the other three models. In terms the width of the transition region from complete to incomplete quantified by σ_c^{pred} , GR_{AE} consistently achieves values ranging 0.3 to 0.4 across all six study regions (Table 1; Figure S4). GR_{SS} and GR_{CO} give similar results, while GR_{BS} yields larger estimates of σ_c^{pred} . However, when compared to the other models, these elevated values appear to reflect not a genuine structure of the seismic catalog, but rather a reduced ability of GR_{BS} to accurately and precisely model the censored empirical distribution of earthquake magnitudes.

Table S5(a) also reports the Rmse, R-Square, SSE, and AIC performance metrics for the augmented Gutenberg-Richter models, while Table S5(b) provides the corresponding rankings of the augmented Gutenberg-Richter models based on these metrics. These rankings are also visually summarized through pie charts in Figure 3. It can be observed that GR_{BS} tends to exhibit polarized performance, frequently ranking either as the best or the worst across evaluations. In contrast, GR_{SS} and GR_{AE} tend to occupy intermediate performance ranks, with GR_{AE} standing out by attaining best or good rankings significantly more often than the other three models.

Table S5 also report the p -values and the corresponding model rankings. The p -values were computed following these steps: (1) Fit an empirical catalog to obtain the best-fitting model (Figures 4 & S6 to S8) and calculate the Kolmogorov-Smirnov (KS) statistic of the observed catalog relative to the best-fitting model, denoted as KS_{obs} ; (2) Generate 10,000 sets of simulated catalogs using the best-fitting model, and compute the KS statistic for each, denoted as KS_{sim}^i ($i = 1, 2, \dots, 10,000$); (3) Calculate the p -value as the proportion of KS_{sim}^i that are larger than KS_{obs} . To simulate randomness within the synthetic earthquake catalogs, a uniform random noise uniformly sampled in the interval $[-0.5, 0.5]$ was added to each synthetic magnitude. According to the p -value rankings, GR_{SS} and GR_{AE} perform the best, being ranked as best or good in 100% of cases. GR_{BS} performs comparably and fall in the middle range, each being ranked as best in about 67% of cases. GR_{CO} performs the worst, receiving a best ranking in 33.3% of the cases but a worst ranking in 33.3% as well.

Text S4. GR_{AE} results on empirical catalog

The Rmse values for GR_{AE} vary between approximately 4.0×10^2 and 6.0×10^3 across the six regions. The highest Rmse is observed in Sichuan-Yunnan $[(5.79 \pm 0.34) \times 10^3]$ and California $[(4.50 \pm 1.04) \times 10^3]$, while the lowest occurs in northern Xinjiang $[(3.96 \pm 1.16) \times 10^2]$ and Southeastern coastal $[(4.59 \pm 1.13) \times 10^2]$. The *R*-Square values are above 0.9985 in all regions, indicating excellent model fit. The SSE and AIC metrics, however, depend on dataset size and thus are not directly comparable across the six study areas. The GR_{AE} model demonstrates consistently strong performance across diverse regions, with notably high *p*-values (≥ 0.8942) indicating excellent fits for northern Xinjiang, California, southeastern coastal of China, and Sichuan-Yunnan regions. Moderate results in New Zealand (*p*-value = 0.5240) and Beijing-Tianjin-Hebei (*p*-value = 0.3050) highlight regional complexities, affirming the model's flexibility and robustness in capturing realistic earthquake completeness and magnitude distributions.

For six regions, the uncertainties in the estimated a^{pred} and b^{pred} -values remain below 0.14 and 0.06, respectively. In terms of the mean b^{pred} -values, Sichuan-Yunnan (0.87 ± 0.01) yields values below 1.0, while northern Xinjiang (0.92 ± 0.06), the southeastern coast (0.96 ± 0.05) and Beijing-Tianjin-Hebei (0.97 ± 0.04) are closest to 1. In contrast, California (1.17 ± 0.04) and New Zealand (1.11 ± 0.02) exhibit mean b^{pred} -values greater than 1.

For m_c^{pred} , the uncertainty in its parameter estimates across all regions is below 0.16, and the uncertainty in σ_c^{pred} is consistently less than 0.06, reflecting the robustness of these estimates. As outlined in Section 2, σ_c quantifies the intrinsic smooth transition from incompleteness to completeness as magnitudes crossover from below to above m_c and should not be confused with the standard error of m_c^{pred} reported in Table 1. Based on the GR_{AE} results, the estimated m_c^{pred} and σ_c^{pred} values are approximately 0.76 and 0.38 for Beijing-Tianjin-Hebei, 0.58 and 0.32 for the southeastern coast, 1.09 and 0.39 for Sichuan-Yunnan, 1.47 and 0.39 for northern Xinjiang, 0.96 and 0.41 for California, and 2.07 and 0.39 for New Zealand. These results highlight the capacity of GR_{AE} to provide not only stable parameter estimates but also a probabilistic description of catalog completeness, which can guide users in selecting magnitude thresholds based on the desired level of completeness.

Table S1(a). Fitted parameters and performance evaluation metrics of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs. Each catalog is generated by the GR_{SS} model using predefined parameters. Among these parameters, the true a -value is calculated as $a^{\text{true}} = k + 2.5 + 0.75\text{gReLU}(-10/3)$, based on Equations (1) and (2). Light red shading indicates scenarios with correct model specification; light blue shading indicates scenarios with model mis-specification.

Model	Predefined parameters		Fitted parameters					Rmse	R-Square	SSE	AIC
	θ^{true}	$N^{\text{true}} = 10^k$	a^{pred}	b^{pred}	m_c^{pred}	σ_c^{pred}	ζ^{pred}				
GR _{SS}	$b^{\text{true}} = 1.00$ $m_c^{\text{true}} = 2.50$ $\sigma_c^{\text{true}} = 0.75$	$k = 3$	5.35 ± 0.62	0.94 ± 0.15	2.45 ± 0.26	0.66 ± 0.08	--	$(1.25 \pm 0.34) \times 10^1$	0.9988 ± 0.0007	$(9.83 \pm 5.51) \times 10^3$	$(3.05 \pm 0.36) \times 10^2$
		$k = 4$	6.53 ± 0.42	0.98 ± 0.09	2.55 ± 0.18	0.69 ± 0.06	--	$(1.19 \pm 0.33) \times 10^2$	0.9989 ± 0.0006	$(1.02 \pm 0.56) \times 10^6$	$(6.58 \pm 0.45) \times 10^2$
		$k = 5$	7.60 ± 0.24	0.99 ± 0.05	2.60 ± 0.11	0.71 ± 0.04	--	$(1.15 \pm 0.32) \times 10^3$	0.9989 ± 0.0006	$(1.07 \pm 0.62) \times 10^8$	$(1.08 \pm 0.04) \times 10^3$
		$k = 6$	8.72 ± 0.12	1.00 ± 0.02	2.66 ± 0.06	0.74 ± 0.02	--	$(1.31 \pm 0.23) \times 10^4$	0.9986 ± 0.0005	$(1.43 \pm 0.51) \times 10^{10}$	$(1.53 \pm 0.03) \times 10^3$
GR _{BS}	--	$k = 3$	6.95 ± 1.25	1.15 ± 0.20	2.84 ± 0.61	0.55 ± 0.07	--	$(2.74 \pm 0.64) \times 10^1$	0.9943 ± 0.0027	$(4.65 \pm 2.05) \times 10^4$	$(4.00 \pm 0.39) \times 10^2$
		$k = 4$	7.96 ± 1.04	1.17 ± 0.16	2.82 ± 0.50	0.54 ± 0.07	--	$(2.70 \pm 0.49) \times 10^2$	0.9945 ± 0.0021	$(5.13 \pm 2.00) \times 10^6$	$(7.74 \pm 0.57) \times 10^2$
		$k = 5$	8.86 ± 0.84	1.16 ± 0.12	2.80 ± 0.40	0.54 ± 0.05	--	$(2.52 \pm 0.38) \times 10^3$	0.9951 ± 0.0015	$(4.98 \pm 1.63) \times 10^8$	$(1.20 \pm 0.06) \times 10^3$
		$k = 6$	9.93 ± 0.52	1.17 ± 0.07	2.85 ± 0.25	0.55 ± 0.03	--	$(2.31 \pm 0.22) \times 10^4$	0.9958 ± 0.0008	$(4.33 \pm 0.84) \times 10^{10}$	$(1.63 \pm 0.03) \times 10^3$
GR _{CO}	--	$k = 3$	4.78 ± 0.33	0.82 ± 0.09	2.14 ± 0.15	0.95 ± 0.07	--	$(2.67 \pm 0.60) \times 10^1$	0.9946 ± 0.0026	$(4.42 \pm 2.02) \times 10^4$	$(3.98 \pm 0.36) \times 10^2$
		$k = 4$	6.02 ± 0.20	0.89 ± 0.05	2.27 ± 0.09	1.01 ± 0.05	--	$(2.98 \pm 0.47) \times 10^2$	0.9933 ± 0.0020	$(6.14 \pm 1.68) \times 10^6$	$(7.87 \pm 0.45) \times 10^2$
		$k = 5$	7.18 ± 0.11	0.93 ± 0.03	2.35 ± 0.05	1.06 ± 0.03	--	$(3.27 \pm 0.31) \times 10^3$	0.9918 ± 0.0015	$(8.22 \pm 1.42) \times 10^8$	$(1.24 \pm 0.04) \times 10^3$
		$k = 6$	8.32 ± 0.06	0.96 ± 0.01	2.42 ± 0.03	1.11 ± 0.02	--	$(3.63 \pm 0.17) \times 10^4$	0.9898 ± 0.0010	$(1.06 \pm 0.10) \times 10^{11}$	$(1.70 \pm 0.02) \times 10^3$
GR _{AE}	--	$k = 3$	5.32 ± 0.50	0.92 ± 0.13	2.50 ± 0.23	0.86 ± 0.06	0.86 ± 0.29	$(1.94 \pm 0.47) \times 10^1$	0.9971 ± 0.0015	$(2.33 \pm 1.11) \times 10^4$	$(3.61 \pm 0.34) \times 10^2$
		$k = 4$	6.57 ± 0.43	0.98 ± 0.09	2.59 ± 0.20	0.90 ± 0.05	0.89 ± 0.19	$(2.02 \pm 0.40) \times 10^2$	0.9969 ± 0.0012	$(2.86 \pm 1.03) \times 10^6$	$(7.35 \pm 0.42) \times 10^2$
		$k = 5$	7.75 ± 0.30	1.02 ± 0.05	2.69 ± 0.15	0.92 ± 0.04	0.79 ± 0.12	$(2.05 \pm 0.32) \times 10^3$	0.9967 ± 0.0010	$(3.26 \pm 0.99) \times 10^8$	$(1.17 \pm 0.04) \times 10^3$
		$k = 6$	8.98 ± 0.16	1.05 ± 0.03	2.82 ± 0.08	0.95 ± 0.03	0.67 ± 0.04	$(2.20 \pm 0.22) \times 10^4$	0.9962 ± 0.0008	$(3.92 \pm 0.79) \times 10^{10}$	$(1.62 \pm 0.02) \times 10^3$

Table S1(b). Performance ranking of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs, based on the evaluation metrics shown in Table S1(a).

Metrics	Predefined parameters	Rank			
	$N^{\text{true}} = 10^k$	First	Second	Third	Fourth
Rmse	$k = 3$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 5$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
R-Square	$k = 3$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 5$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
SSE	$k = 3$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 5$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
AIC	$k = 3$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 5$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{BS}	GR _{CO}

Table S2(a). Fitted parameters and performance evaluation metrics of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs. Each catalog is generated by the GR_{BS} model using predefined parameters. The details are same as Table S1(a).

Model	Predefined parameters		Fitted parameters					Rmse	R-Square	SSE	AIC
	θ^{true}	$N^{\text{true}} = 10^k$	a^{pred}	b^{pred}	m_c^{pred}	σ_c^{pred}	ζ^{pred}				
GR _{SS}	--	$k = 3$	4.40 ± 0.28	0.80 ± 0.09	1.74 ± 0.15	0.44 ± 0.06	--	$(1.91 \pm 0.60) \times 10^1$	0.9970 ± 0.0018	$(2.10 \pm 1.20) \times 10^4$	$(3.20 \pm 0.37) \times 10^2$
		$k = 4$	5.58 ± 0.20	0.85 ± 0.05	1.86 ± 0.11	0.51 ± 0.05	--	$(2.36 \pm 0.61) \times 10^2$	0.9955 ± 0.0021	$(3.78 \pm 1.75) \times 10^6$	$(7.13 \pm 0.46) \times 10^2$
		$k = 5$	6.71 ± 0.14	0.88 ± 0.03	1.95 ± 0.08	0.56 ± 0.04	--	$(2.75 \pm 0.50) \times 10^3$	0.9939 ± 0.0021	$(5.75 \pm 1.94) \times 10^8$	$(1.18 \pm 0.05) \times 10^3$
		$k = 6$	7.90 ± 0.08	0.92 ± 0.02	2.06 ± 0.05	0.63 ± 0.03	--	$(3.46 \pm 0.31) \times 10^4$	0.9907 ± 0.0017	$(9.66 \pm 1.72) \times 10^{10}$	$(1.68 \pm 0.03) \times 10^3$
GR _{BS}	$b^{\text{true}} = 1.00$ $m_c^{\text{true}} = 2.50$ $\sigma_c^{\text{true}} = 0.75$	$k = 3$	5.27 ± 1.18	0.94 ± 0.21	1.98 ± 0.79	0.53 ± 0.16	--	$(8.00 \pm 3.18) \times 10^0$	0.9994 ± 0.0005	$(3.99 \pm 3.51) \times 10^3$	$(2.55 \pm 0.47) \times 10^2$
		$k = 4$	6.59 ± 1.19	0.99 ± 0.17	2.27 ± 0.87	0.60 ± 0.17	--	$(5.13 \pm 2.79) \times 10^1$	0.9997 ± 0.0003	$(2.22 \pm 2.44) \times 10^5$	$(5.00 \pm 0.85) \times 10^2$
		$k = 5$	7.54 ± 0.84	0.99 ± 0.11	2.33 ± 0.70	0.63 ± 0.15	--	$(4.15 \pm 2.41) \times 10^2$	0.9998 ± 0.0002	$(1.69 \pm 1.97) \times 10^7$	$(8.78 \pm 1.00) \times 10^2$
		$k = 6$	9.13 ± 0.65	1.06 ± 0.07	2.91 ± 0.42	0.77 ± 0.07	--	$(5.62 \pm 2.92) \times 10^3$	0.9997 ± 0.0003	$(3.20 \pm 3.03) \times 10^9$	$(1.36 \pm 0.10) \times 10^3$
GR _{CO}	--	$k = 3$	4.22 ± 0.15	0.76 ± 0.06	1.61 ± 0.09	0.67 ± 0.06	--	$(3.31 \pm 0.61) \times 10^1$	0.9914 ± 0.0031	$(5.99 \pm 2.05) \times 10^4$	$(3.83 \pm 0.35) \times 10^2$
		$k = 4$	5.38 ± 0.11	0.80 ± 0.03	1.71 ± 0.07	0.75 ± 0.05	--	$(3.78 \pm 0.52) \times 10^2$	0.9889 ± 0.0028	$(9.31 \pm 2.23) \times 10^6$	$(7.77 \pm 0.50) \times 10^2$
		$k = 5$	6.51 ± 0.08	0.84 ± 0.02	1.80 ± 0.05	0.83 ± 0.04	--	$(4.26 \pm 0.39) \times 10^3$	0.9856 ± 0.0025	$(1.35 \pm 0.22) \times 10^9$	$(1.25 \pm 0.05) \times 10^3$
		$k = 6$	7.66 ± 0.05	0.88 ± 0.01	1.89 ± 0.03	0.92 ± 0.02	--	$(4.94 \pm 0.22) \times 10^4$	0.9812 ± 0.0017	$(1.96 \pm 0.17) \times 10^{11}$	$(1.74 \pm 0.02) \times 10^3$
GR _{AE}	--	$k = 3$	4.48 ± 0.26	0.81 ± 0.08	1.82 ± 0.15	0.56 ± 0.07	0.67 ± 0.22	$(2.21 \pm 0.58) \times 10^1$	0.9960 ± 0.0020	$(2.76 \pm 1.34) \times 10^4$	$(3.40 \pm 0.34) \times 10^2$
		$k = 4$	5.70 ± 0.22	0.87 ± 0.05	1.95 ± 0.13	0.63 ± 0.06	0.65 ± 0.09	$(2.57 \pm 0.57) \times 10^2$	0.9947 ± 0.0022	$(4.42 \pm 1.76) \times 10^6$	$(7.27 \pm 0.46) \times 10^2$
		$k = 5$	6.87 ± 0.18	0.90 ± 0.04	2.07 ± 0.11	0.68 ± 0.05	0.59 ± 0.05	$(2.87 \pm 0.48) \times 10^3$	0.9933 ± 0.0021	$(6.23 \pm 1.94) \times 10^8$	$(1.19 \pm 0.05) \times 10^3$
		$k = 6$	8.13 ± 0.10	0.95 ± 0.02	2.24 ± 0.06	0.76 ± 0.03	0.54 ± 0.01	$(3.52 \pm 0.30) \times 10^4$	0.9904 ± 0.0016	$(9.95 \pm 1.68) \times 10^{10}$	$(1.68 \pm 0.03) \times 10^3$

Table S2(b). Performance ranking of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs, based on the evaluation metrics shown in Table S2(a).

Metrics	Predefined parameters	Rank			
	$N^{\text{true}} = 10^k$	First	Second	Third	Fourth
Rmse	$k = 3$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 4$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 5$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 6$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
R-Square	$k = 3$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 4$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 5$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 6$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
SSE	$k = 3$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 4$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 5$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 6$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
AIC	$k = 3$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 4$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 5$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	$k = 6$	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}

Table S3(a). Fitted parameters and performance evaluation metrics of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs. Each catalog is generated by the GR_{CO} model using predefined parameters. The details are same as Table S1(a).

Model	Predefined parameters		Fitted parameters					Rmse	R-Square	SSE	AIC
	θ^{true}	$N^{\text{true}} = 10^k$	a^{pred}	b^{pred}	m_c^{pred}	σ_c^{pred}	ζ^{pred}				
GR _{SS}	--	$k = 3$	6.02 ± 0.53	1.08 ± 0.13	2.78 ± 0.16	0.48 ± 0.06	--	$(1.86 \pm 0.66) \times 10^1$	0.9976 ± 0.0016	$(2.33 \pm 1.62) \times 10^4$	$(3.49 \pm 0.62) \times 10^2$
		$k = 4$	6.88 ± 0.32	1.05 ± 0.08	2.73 ± 0.11	0.46 ± 0.04	--	$(2.02 \pm 0.51) \times 10^2$	0.9973 ± 0.0013	$(3.01 \pm 1.56) \times 10^6$	$(7.32 \pm 0.73) \times 10^2$
		$k = 5$	7.79 ± 0.17	1.03 ± 0.04	2.70 ± 0.06	0.44 ± 0.03	--	$(2.05 \pm 0.38) \times 10^3$	0.9973 ± 0.0010	$(3.32 \pm 1.25) \times 10^8$	$(1.17 \pm 0.06) \times 10^3$
		$k = 6$	8.81 ± 0.09	1.04 ± 0.02	2.71 ± 0.04	0.45 ± 0.02	--	$(1.93 \pm 0.26) \times 10^4$	0.9976 ± 0.0006	$(3.06 \pm 0.80) \times 10^{10}$	$(1.60 \pm 0.03) \times 10^3$
GR _{BS}	--	$k = 3$	6.50 ± 0.59	1.16 ± 0.13	2.61 ± 0.28	0.33 ± 0.06	--	$(3.61 \pm 0.55) \times 10^1$	0.9917 ± 0.0028	$(7.81 \pm 2.29) \times 10^4$	$(4.32 \pm 0.38) \times 10^2$
		$k = 4$	7.30 ± 0.37	1.13 ± 0.08	2.59 ± 0.26	0.32 ± 0.05	--	$(3.45 \pm 0.27) \times 10^2$	0.9927 ± 0.0012	$(8.13 \pm 1.22) \times 10^6$	$(8.08 \pm 0.55) \times 10^2$
		$k = 5$	8.17 ± 0.18	1.10 ± 0.04	2.68 ± 0.27	0.34 ± 0.05	--	$(3.26 \pm 0.17) \times 10^3$	0.9934 ± 0.0007	$(8.09 \pm 0.78) \times 10^8$	$(1.24 \pm 0.05) \times 10^3$
		$k = 6$	9.15 ± 0.08	1.09 ± 0.02	2.79 ± 0.21	0.35 ± 0.03	--	$(3.16 \pm 0.09) \times 10^4$	0.9937 ± 0.0004	$(8.05 \pm 0.46) \times 10^{10}$	$(1.68 \pm 0.02) \times 10^3$
GR _{CO}	$b^{\text{true}} = 1.00$ $m_c^{\text{true}} = 2.50$ $\sigma_c^{\text{true}} = 0.75$	$k = 3$	5.71 ± 0.41	1.02 ± 0.11	2.64 ± 0.13	0.76 ± 0.07	--	$(1.28 \pm 0.81) \times 10^1$	0.9986 ± 0.0024	$(1.37 \pm 2.33) \times 10^4$	$(2.95 \pm 0.66) \times 10^2$
		$k = 4$	6.68 ± 0.20	1.02 ± 0.05	2.64 ± 0.07	0.74 ± 0.04	--	$(8.69 \pm 4.46) \times 10^1$	0.9994 ± 0.0007	$(6.50 \pm 8.26) \times 10^5$	$(6.06 \pm 0.70) \times 10^2$
		$k = 5$	7.66 ± 0.11	1.01 ± 0.03	2.63 ± 0.04	0.74 ± 0.03	--	$(7.48 \pm 3.53) \times 10^2$	0.9996 ± 0.0005	$(5.19 \pm 5.66) \times 10^7$	$(1.00 \pm 0.07) \times 10^3$
		$k = 6$	8.67 ± 0.06	1.01 ± 0.01	2.63 ± 0.02	0.74 ± 0.01	--	$(6.25 \pm 1.46) \times 10^3$	0.9997 ± 0.0002	$(3.32 \pm 1.95) \times 10^9$	$(1.41 \pm 0.03) \times 10^3$
GR _{AE}	--	$k = 3$	6.05 ± 0.61	1.05 ± 0.15	2.90 ± 0.29	0.72 ± 0.04	1.05 ± 0.85	$(9.06 \pm 3.01) \times 10^0$	0.9994 ± 0.0004	$(5.50 \pm 4.30) \times 10^3$	$(2.68 \pm 0.50) \times 10^2$
		$k = 4$	6.86 ± 0.44	1.05 ± 0.09	2.72 ± 0.18	0.74 ± 0.04	1.59 ± 0.57	$(7.94 \pm 2.51) \times 10^1$	0.9996 ± 0.0003	$(4.80 \pm 3.34) \times 10^5$	$(6.04 \pm 0.69) \times 10^2$
		$k = 5$	7.76 ± 0.28	1.03 ± 0.05	2.68 ± 0.12	0.74 ± 0.04	1.67 ± 0.43	$(7.57 \pm 2.09) \times 10^2$	0.9996 ± 0.0002	$(4.72 \pm 2.84) \times 10^7$	$(1.02 \pm 0.07) \times 10^3$
		$k = 6$	8.79 ± 0.23	1.03 ± 0.03	2.70 ± 0.12	0.73 ± 0.03	1.58 ± 0.37	$(6.48 \pm 1.32) \times 10^3$	0.9997 ± 0.0001	$(3.52 \pm 1.48) \times 10^9$	$(1.42 \pm 0.04) \times 10^3$

Table S3(b). Performance ranking of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs, based on the evaluation metrics shown in Table S3(a).

Metrics	Predefined parameters	Rank			
	$N^{\text{true}} = 10^k$	First	Second	Third	Fourth
Rmse	$k = 3$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 5$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}
	$k = 6$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}
R-Square	$k = 3$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 6$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}
SSE	$k = 3$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 6$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}
AIC	$k = 3$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	$k = 5$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}
	$k = 6$	GR _{CO}	GR _{AE}	GR _{SS}	GR _{BS}

Table S4(a). Fitted parameters and performance evaluation metrics of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to simulated catalogs. Each catalog is generated by the GR_{AE} model using predefined parameters. The details are same as Table S1(a).

Model	Predefined parameters		Fitted parameters					Rmse	R-Square	SSE	AIC
	θ^{true}	$N^{\text{true}} = 10^k$	a^{pred}	b^{pred}	m_c^{pred}	σ_c^{pred}	ζ^{pred}				
GR _{SS}	--	$k = 3$	4.98 ± 0.39	0.91 ± 0.11	2.17 ± 0.16	0.50 ± 0.06	--	$(1.27 \pm 0.53) \times 10^1$	0.9987 ± 0.0012	$(1.12 \pm 1.08) \times 10^4$	$(2.99 \pm 0.62) \times 10^2$
		$k = 4$	5.97 ± 0.20	0.91 ± 0.05	2.17 ± 0.09	0.49 ± 0.04	--	$(9.96 \pm 3.23) \times 10^1$	0.9992 ± 0.0005	$(7.53 \pm 5.67) \times 10^5$	$(6.27 \pm 0.69) \times 10^2$
		$k = 5$	7.03 ± 0.13	0.92 ± 0.03	2.20 ± 0.07	0.51 ± 0.03	--	$(8.02 \pm 1.82) \times 10^2$	0.9995 ± 0.0003	$(5.15 \pm 2.83) \times 10^7$	$(1.02 \pm 0.05) \times 10^3$
		$k = 6$	8.15 ± 0.08	0.95 ± 0.02	2.26 ± 0.04	0.55 ± 0.02	--	$(7.82 \pm 1.61) \times 10^3$	0.9995 ± 0.0002	$(5.10 \pm 2.30) \times 10^9$	$(1.44 \pm 0.03) \times 10^3$
GR _{BS}	--	$k = 3$	5.58 ± 0.70	1.01 ± 0.16	2.14 ± 0.32	0.44 ± 0.08	--	$(2.80 \pm 0.45) \times 10^1$	0.9946 ± 0.0017	$(4.69 \pm 1.52) \times 10^4$	$(3.99 \pm 0.44) \times 10^2$
		$k = 4$	6.40 ± 0.43	0.98 ± 0.09	2.09 ± 0.28	0.43 ± 0.07	--	$(2.59 \pm 0.25) \times 10^2$	0.9954 ± 0.0009	$(4.59 \pm 0.94) \times 10^6$	$(7.61 \pm 0.54) \times 10^2$
		$k = 5$	7.39 ± 0.33	0.98 ± 0.06	2.10 ± 0.31	0.43 ± 0.06	--	$(2.43 \pm 0.16) \times 10^3$	0.9958 ± 0.0005	$(4.51 \pm 0.62) \times 10^8$	$(1.19 \pm 0.05) \times 10^3$
		$k = 6$	8.50 ± 0.25	1.00 ± 0.04	2.20 ± 0.34	0.46 ± 0.07	--	$(2.31 \pm 0.10) \times 10^4$	0.9961 ± 0.0003	$(4.28 \pm 0.36) \times 10^{10}$	$(1.62 \pm 0.02) \times 10^3$
GR _{CO}	--	$k = 3$	4.79 ± 0.26	0.87 ± 0.08	2.06 ± 0.12	0.78 ± 0.08	--	$(1.99 \pm 0.98) \times 10^1$	0.9967 ± 0.0035	$(2.91 \pm 3.13) \times 10^4$	$(3.48 \pm 0.68) \times 10^2$
		$k = 4$	5.83 ± 0.14	0.88 ± 0.04	2.08 ± 0.07	0.79 ± 0.05	--	$(1.82 \pm 0.72) \times 10^2$	0.9974 ± 0.0021	$(2.59 \pm 2.14) \times 10^6$	$(7.04 \pm 0.66) \times 10^2$
		$k = 5$	6.88 ± 0.08	0.90 ± 0.02	2.10 ± 0.04	0.80 ± 0.03	--	$(1.79 \pm 0.41) \times 10^3$	0.9976 ± 0.0012	$(2.56 \pm 1.30) \times 10^8$	$(1.14 \pm 0.05) \times 10^3$
		$k = 6$	7.95 ± 0.05	0.91 ± 0.01	2.13 ± 0.02	0.83 ± 0.02	--	$(1.98 \pm 0.19) \times 10^4$	0.9971 ± 0.0005	$(3.17 \pm 0.59) \times 10^{10}$	$(1.60 \pm 0.02) \times 10^3$
GR _{AE}	$b^{\text{true}} = 1.00$	$k = 3$	5.12 ± 0.39	0.89 ± 0.12	2.36 ± 0.22	0.69 ± 0.03	0.68 ± 0.62	$(1.01 \pm 0.23) \times 10^1$	0.9993 ± 0.0003	$(6.27 \pm 2.88) \times 10^3$	$(2.80 \pm 0.36) \times 10^2$
	$m_c^{\text{true}} = 2.50$	$k = 4$	6.12 ± 0.33	0.92 ± 0.08	2.28 ± 0.21	0.68 ± 0.03	0.88 ± 0.47	$(7.91 \pm 1.39) \times 10^1$	0.9996 ± 0.0002	$(4.35 \pm 1.64) \times 10^5$	$(6.01 \pm 0.43) \times 10^2$
	$\sigma_c^{\text{true}} = 0.75$	$k = 5$	7.27 ± 0.30	0.96 ± 0.05	2.36 ± 0.18	0.68 ± 0.03	0.70 ± 0.31	$(7.28 \pm 1.57) \times 10^2$	0.9996 ± 0.0002	$(4.17 \pm 2.11) \times 10^7$	$(1.01 \pm 0.04) \times 10^3$
	$\zeta^{\text{true}} = 0.35$	$k = 6$	8.55 ± 0.13	1.00 ± 0.02	2.54 ± 0.08	0.71 ± 0.03	0.44 ± 0.10	$(8.64 \pm 1.78) \times 10^3$	0.9994 ± 0.0002	$(6.23 \pm 2.68) \times 10^9$	$(1.46 \pm 0.03) \times 10^3$

Table S4(b). Performance ranking of the augmented Gutenberg-Richter model with four $\text{gReLU}(x)$ functions applied to simulated catalogs, based on the evaluation metrics shown in Table S4(a).

Metrics	Predefined parameters	Rank			
	$N^{\text{true}} = 10^k$	First	Second	Third	Fourth
Rmse	$k = 3$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
R-Square	$k = 3$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
SSE	$k = 3$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
AIC	$k = 3$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 4$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 5$	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
	$k = 6$	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}

Table S5(a). Performance evaluation metrics of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to empirical earthquake catalogs from six study regions.

Model	Region	Rmse	R-Square	SSE	AIC	p -value
GR_{ss}	Beijing-Tianjin-Hebei	$(9.06 \pm 1.62) \times 10^2$	0.9989 ± 0.0004	$(1.30 \pm 0.47) \times 10^8$	$(2.10 \pm 0.05) \times 10^3$	0.2814
	Southeastern coastal	$(4.92 \pm 1.09) \times 10^2$	0.9986 ± 0.0006	$(3.88 \pm 1.71) \times 10^7$	$(1.91 \pm 0.06) \times 10^3$	1.0000
	Sichuan-Yunnan	$(5.54 \pm 0.79) \times 10^3$	0.9996 ± 0.0001	$(5.44 \pm 1.51) \times 10^9$	$(3.00 \pm 0.06) \times 10^3$	0.8650
	Northern Xinjiang	$(6.55 \pm 1.70) \times 10^2$	0.9994 ± 0.0003	$(7.65 \pm 3.89) \times 10^7$	$(2.16 \pm 0.11) \times 10^3$	1.0000
	California	$(5.01 \pm 1.12) \times 10^3$	0.9981 ± 0.0008	$(4.14 \pm 1.64) \times 10^9$	$(2.69 \pm 0.08) \times 10^3$	1.0000
	New Zealand	$(1.03 \pm 0.30) \times 10^3$	0.9999 ± 0.0001	$(1.90 \pm 1.14) \times 10^8$	$(2.29 \pm 0.10) \times 10^3$	0.3859
GR_{BS}	Beijing-Tianjin-Hebei	$(2.64 \pm 0.45) \times 10^2$	0.9999 ± 0.0000	$(1.10 \pm 0.40) \times 10^7$	$(1.72 \pm 0.05) \times 10^3$	0.3461
	Southeastern coastal	$(2.07 \pm 0.36) \times 10^2$	0.9998 ± 0.0001	$(6.79 \pm 2.45) \times 10^6$	$(1.64 \pm 0.05) \times 10^3$	0.9447
	Sichuan-Yunnan	$(1.15 \pm 0.24) \times 10^4$	0.9982 ± 0.0008	$(2.39 \pm 1.02) \times 10^{10}$	$(3.25 \pm 0.08) \times 10^3$	0.5274
	Northern Xinjiang	$(1.14 \pm 0.34) \times 10^3$	0.9980 ± 0.0012	$(2.38 \pm 1.53) \times 10^8$	$(2.35 \pm 0.11) \times 10^3$	1.0000
	California	$(1.36 \pm 0.35) \times 10^3$	0.9999 ± 0.0001	$(3.09 \pm 1.63) \times 10^8$	$(2.28 \pm 0.07) \times 10^3$	1.0000
	New Zealand	$(2.75 \pm 0.71) \times 10^3$	0.9990 ± 0.0005	$(1.34 \pm 0.69) \times 10^9$	$(2.62 \pm 0.09) \times 10^3$	0.5813
GR_{co}	Beijing-Tianjin-Hebei	$(1.46 \pm 0.13) \times 10^3$	0.9973 ± 0.0005	$(3.29 \pm 0.57) \times 10^8$	$(2.25 \pm 0.03) \times 10^3$	0.2408
	Southeastern coastal	$(6.80 \pm 1.07) \times 10^2$	0.9974 ± 0.0008	$(7.26 \pm 2.14) \times 10^7$	$(2.01 \pm 0.04) \times 10^3$	1.0000
	Sichuan-Yunnan	$(5.99 \pm 0.52) \times 10^3$	0.9995 ± 0.0001	$(6.28 \pm 1.17) \times 10^9$	$(3.03 \pm 0.03) \times 10^3$	0.8738
	Northern Xinjiang	$(4.07 \pm 1.44) \times 10^2$	0.9997 ± 0.0002	$(3.12 \pm 2.92) \times 10^7$	$(2.00 \pm 0.11) \times 10^3$	1.0000
	California	$(6.31 \pm 0.85) \times 10^3$	0.9971 ± 0.0008	$(6.39 \pm 1.60) \times 10^9$	$(2.77 \pm 0.06) \times 10^3$	0.7772
	New Zealand	$(1.18 \pm 0.20) \times 10^3$	0.9998 ± 0.0001	$(2.36 \pm 0.79) \times 10^8$	$(2.35 \pm 0.06) \times 10^3$	0.3580
GR_{AE}	Beijing-Tianjin-Hebei	$(9.09 \pm 1.59) \times 10^2$	0.9989 ± 0.0004	$(1.31 \pm 0.49) \times 10^8$	$(2.11 \pm 0.05) \times 10^3$	0.3050
	Southeastern coastal	$(4.59 \pm 1.13) \times 10^2$	0.9988 ± 0.0006	$(3.43 \pm 1.72) \times 10^7$	$(1.89 \pm 0.07) \times 10^3$	0.9961
	Sichuan-Yunnan	$(5.79 \pm 0.34) \times 10^3$	0.9996 ± 0.0001	$(5.85 \pm 0.72) \times 10^9$	$(3.02 \pm 0.03) \times 10^3$	0.8942
	Northern Xinjiang	$(3.96 \pm 1.16) \times 10^2$	0.9998 ± 0.0002	$(2.85 \pm 2.14) \times 10^7$	$(1.99 \pm 0.10) \times 10^3$	1.0000
	California	$(4.50 \pm 1.04) \times 10^3$	0.9985 ± 0.0007	$(3.36 \pm 1.39) \times 10^9$	$(2.66 \pm 0.08) \times 10^3$	1.0000
	New Zealand	$(9.46 \pm 1.14) \times 10^2$	0.9999 ± 0.0000	$(1.50 \pm 0.37) \times 10^8$	$(2.28 \pm 0.04) \times 10^3$	0.5240

Table S5(b). Performance ranking of the augmented Gutenberg-Richter model with four gReLU(x) functions applied to empirical catalogs, based on the evaluation metrics shown in Table S5(a).

Metrics	Region	Rank			
		First	Second	Third	Fourth
Rmse	Beijing-Tianjin-Hebei	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	Southeastern coastal	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	Sichuan-Yunnan	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	Northern Xinjiang	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	California	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	New Zealand	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
R-Square	Beijing-Tianjin-Hebei	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	Southeastern coastal	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	Sichuan-Yunnan	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	Northern Xinjiang	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	California	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	New Zealand	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
SSE	Beijing-Tianjin-Hebei	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	Southeastern coastal	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	Sichuan-Yunnan	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	Northern Xinjiang	GR _{AE}	GR _{BS}	GR _{SS}	GR _{CO}
	California	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	New Zealand	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
AIC	Beijing-Tianjin-Hebei	GR _{BS}	GR _{SS}	GR _{AE}	GR _{CO}
	Southeastern coastal	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	Sichuan-Yunnan	GR _{SS}	GR _{AE}	GR _{CO}	GR _{BS}
	Northern Xinjiang	GR _{AE}	GR _{BS}	GR _{SS}	GR _{CO}
	California	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	New Zealand	GR _{AE}	GR _{SS}	GR _{CO}	GR _{BS}
p -value	Beijing-Tianjin-Hebei	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}
	Southeastern coastal	GR _{SS} = GR _{CO}		GR _{AE}	GR _{BS}
	Sichuan-Yunnan	GR _{AE}	GR _{CO}	GR _{SS}	GR _{BS}
	Northern Xinjiang	GR _{SS} = GR _{BS} = GR _{CO} = GR _{AE}			
	California	GR _{SS} = GR _{BS} = GR _{AE}			GR _{CO}
	New Zealand	GR _{BS}	GR _{AE}	GR _{SS}	GR _{CO}

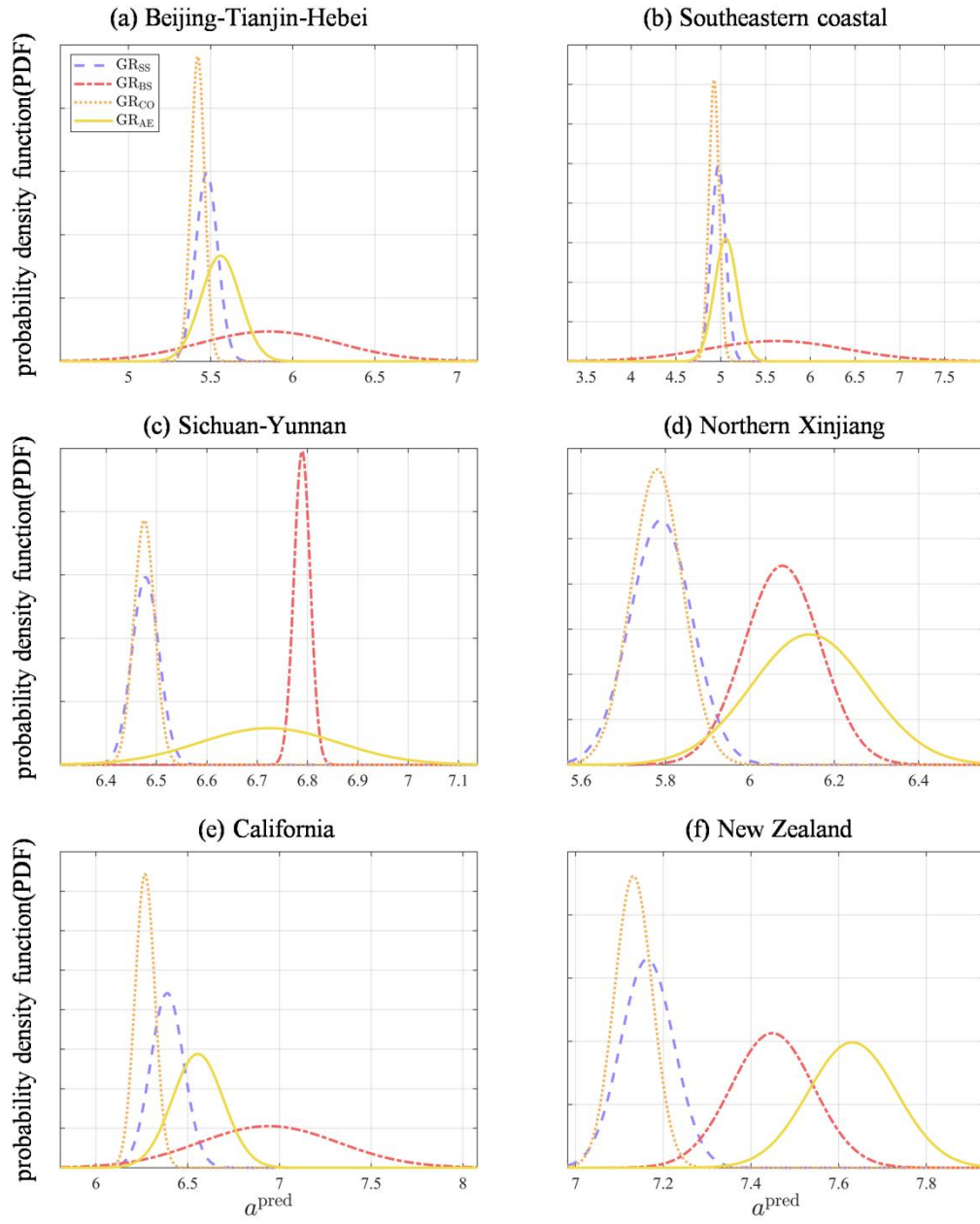


Figure S1. Probability density function (PDF) of the estimated a^{pred} values obtained from 200 bootstrapping iterations using the augmented Gutenberg-Richter law with the four gReLU(x) functions, applied to empirical earthquake catalogs from six study regions.

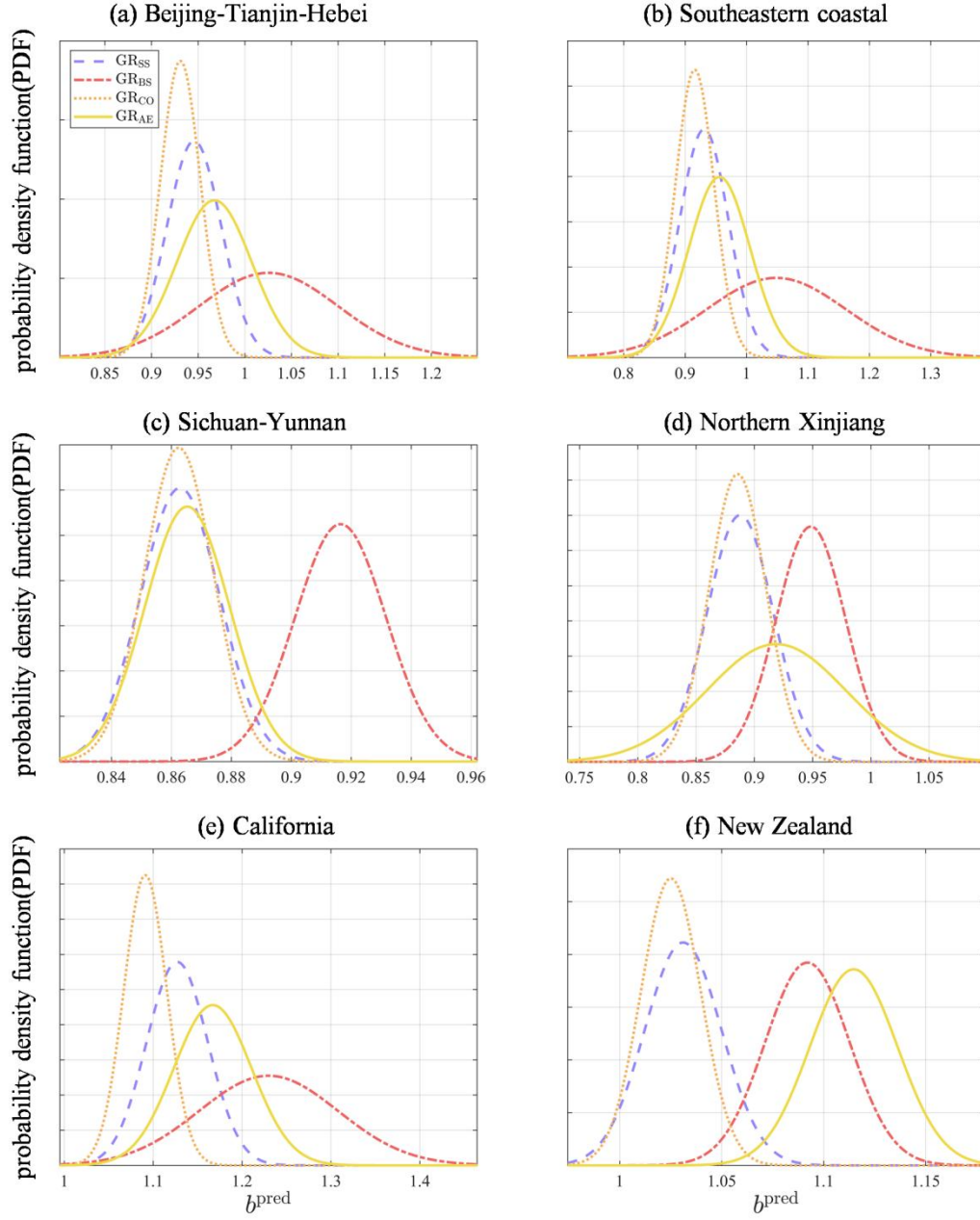


Figure S2. PDF of the estimated b^{pred} values obtained from 200 bootstrapping iterations using the augmented Gutenberg-Richter law with the four gReLU(x) functions, applied to empirical earthquake catalogs from six study regions.

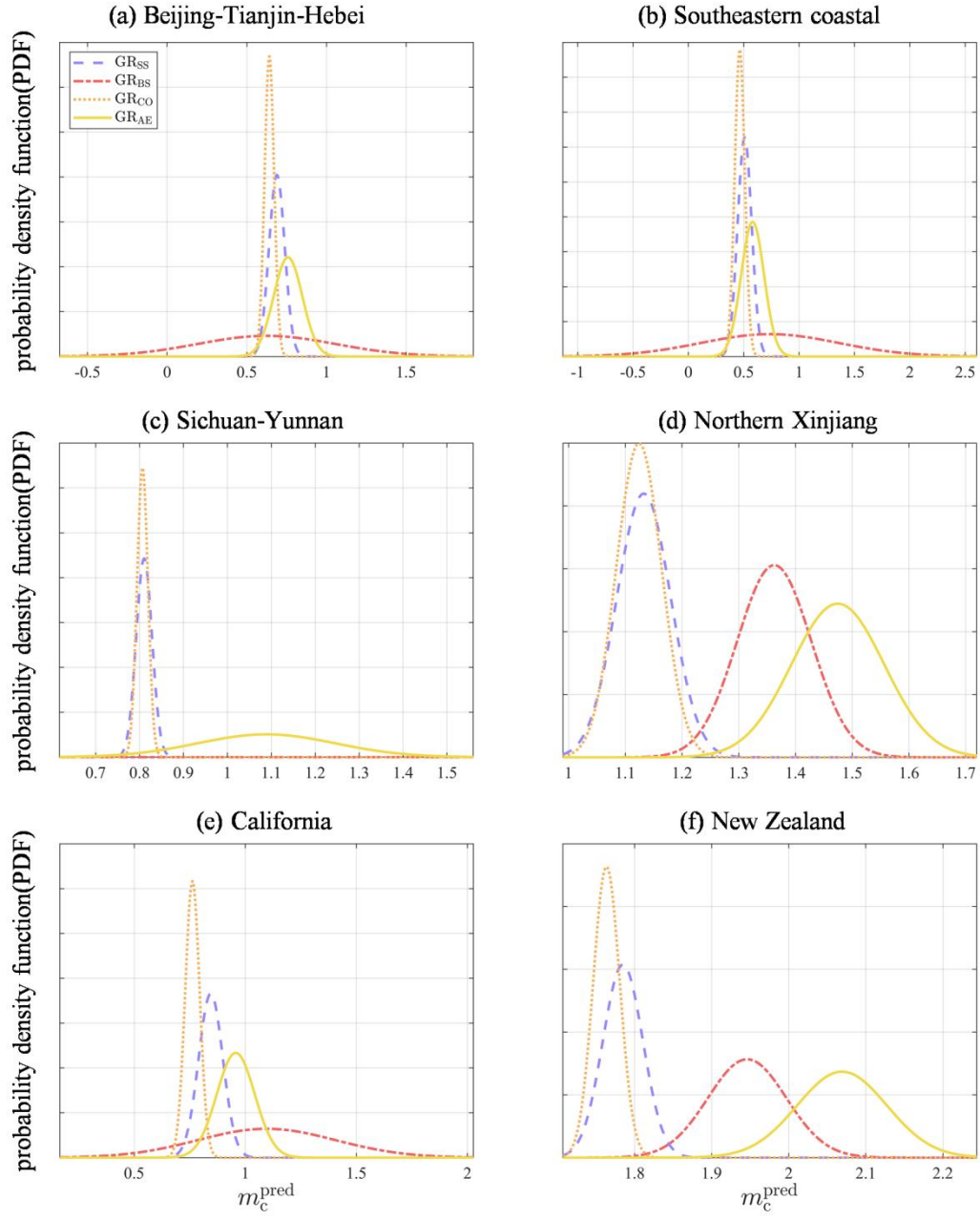


Figure S3. PDF of the estimated m_c^{pred} values obtained from 200 bootstrapping iterations using the augmented Gutenberg-Richter law with the four gReLU(x) functions, applied to observed earthquake catalogs from six study regions.

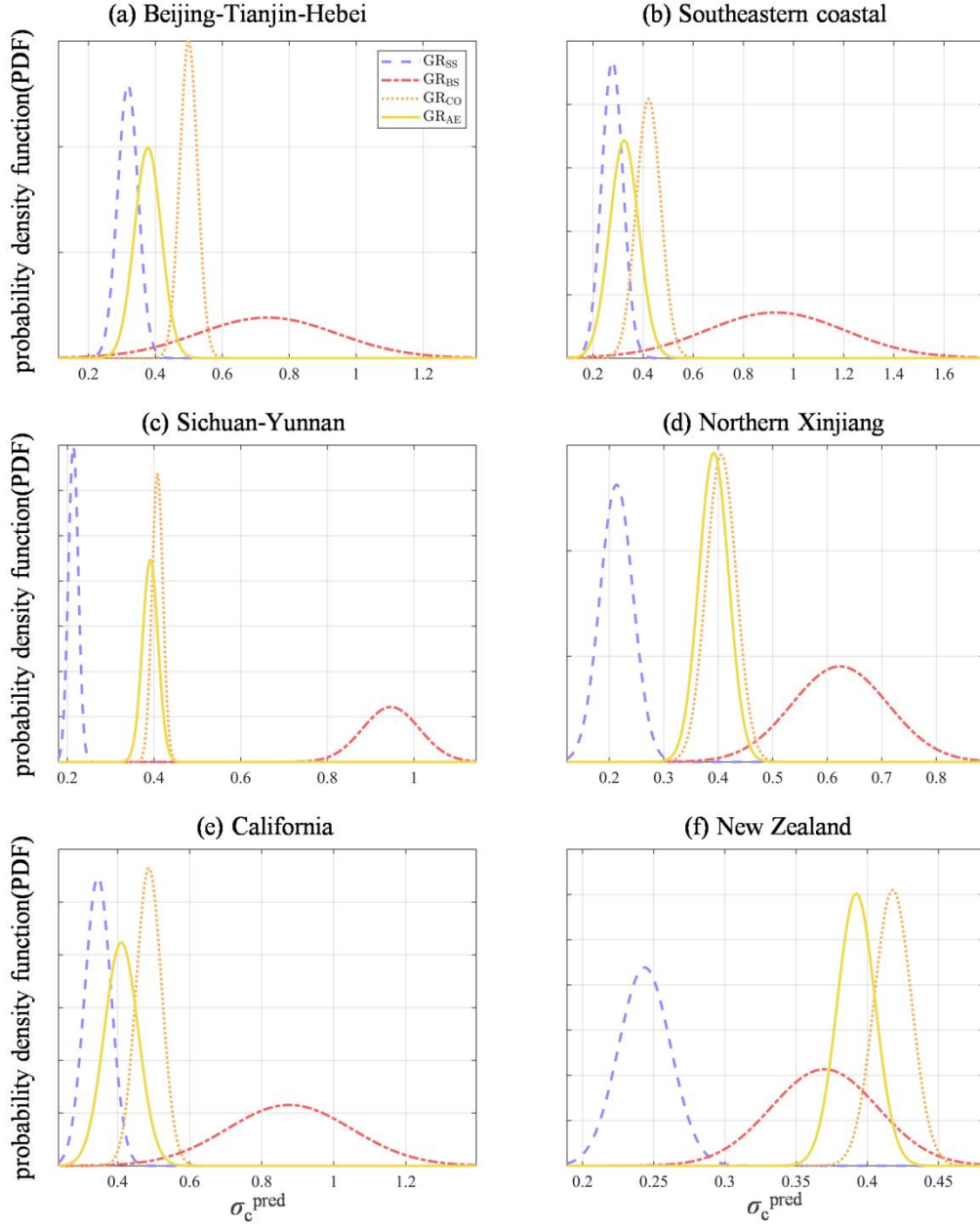


Figure S4. PDF of the estimated σ_c^{pred} values obtained from 200 bootstrapping iterations using the augmented Gutenberg-Richter law with the four gReLU(x) functions, applied to empirical earthquake catalogs from six study regions.

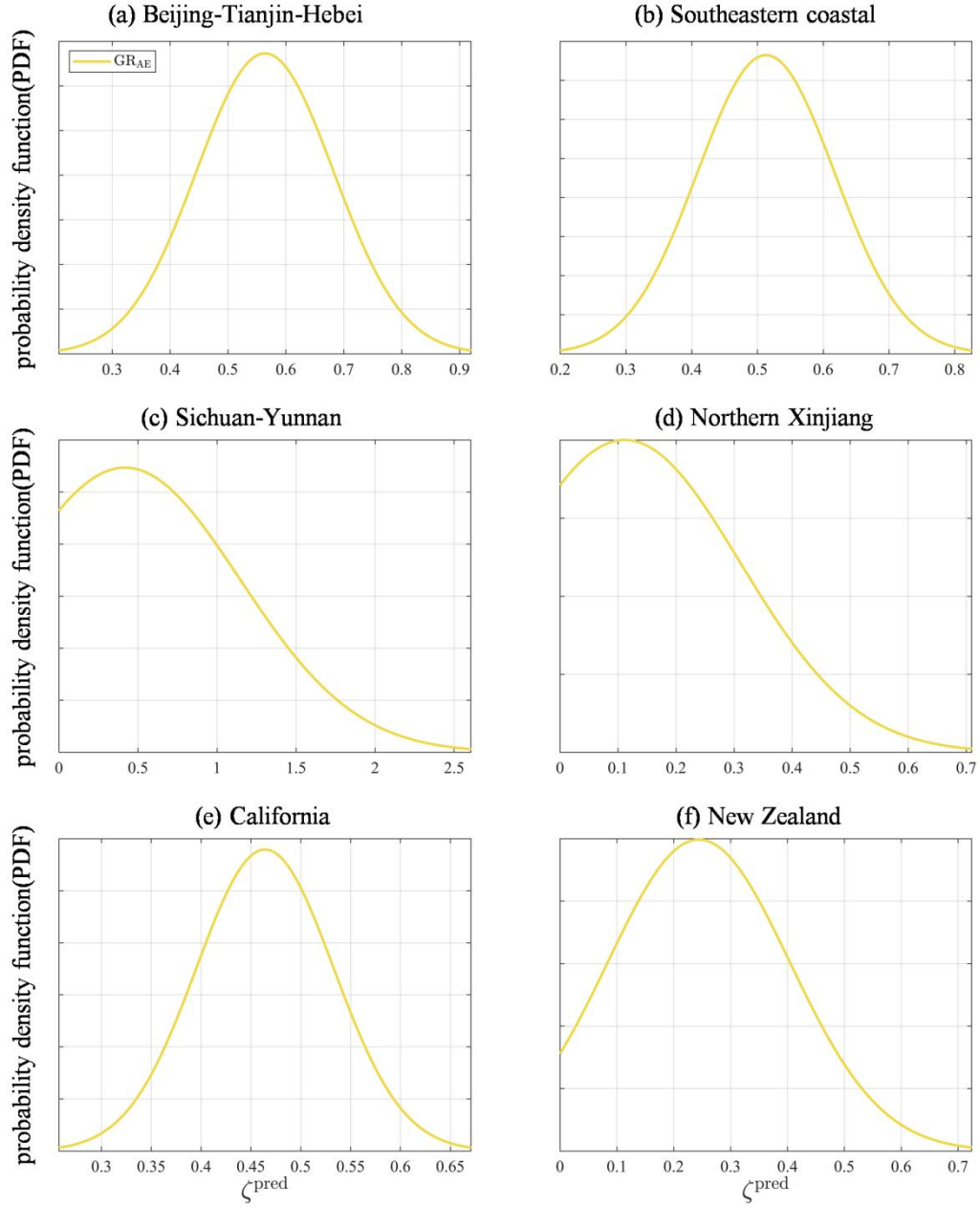


Figure S5. PDF of the estimated ζ^{pred} values obtained from 200 bootstrapping iterations using the augmented Gutenberg-Richter law with the AEReLU(x) function, applied to empirical earthquake catalogs from six study regions.

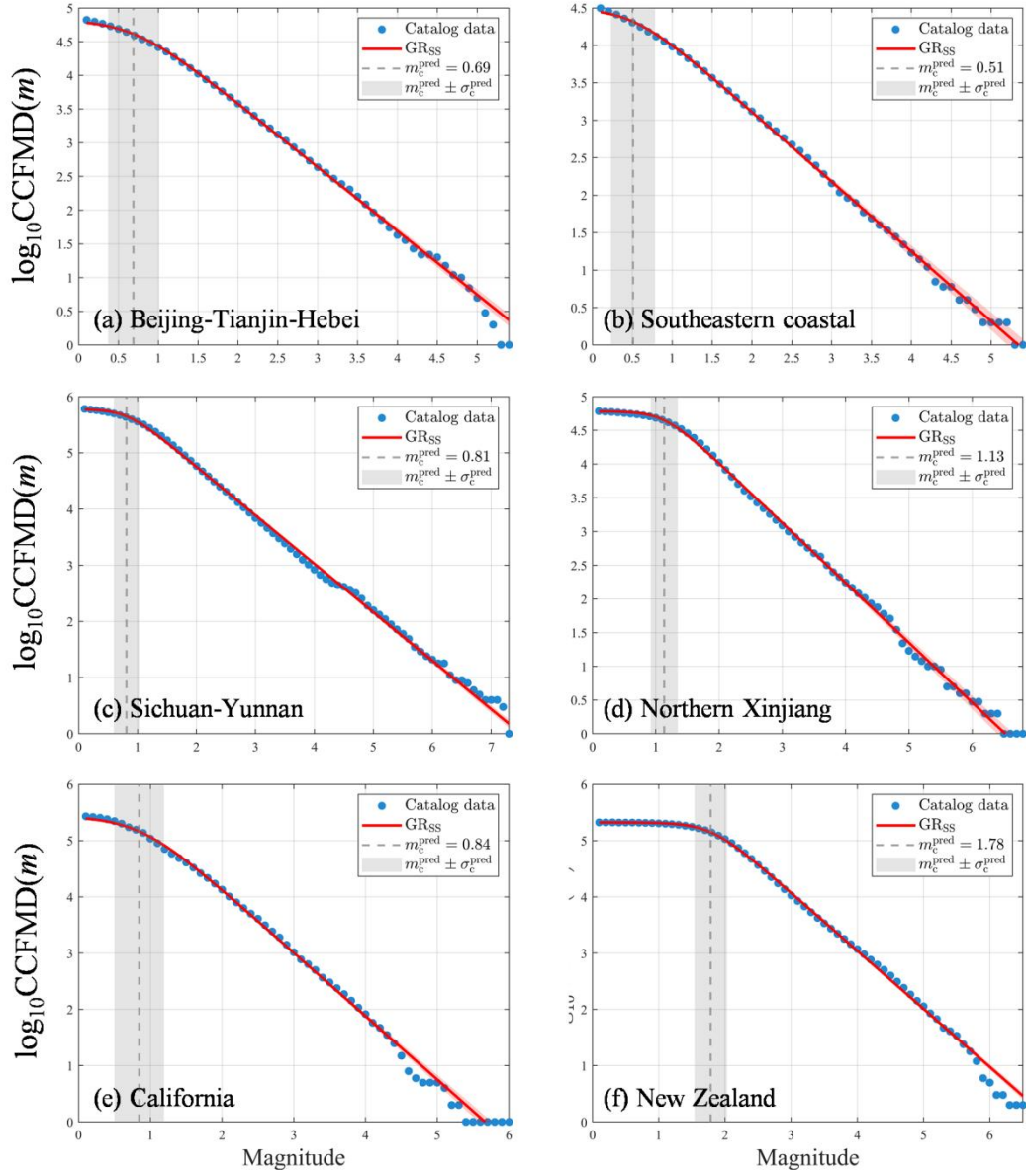


Figure S6. Complementary cumulative frequency-magnitude distribution (CCFMD) of the empirical earthquake catalogs from six study regions, along with the fitted CCFMD(m) derived using the augmented Gutenberg-Richter law with the SSReLU(x) function (GR_{ss}). The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

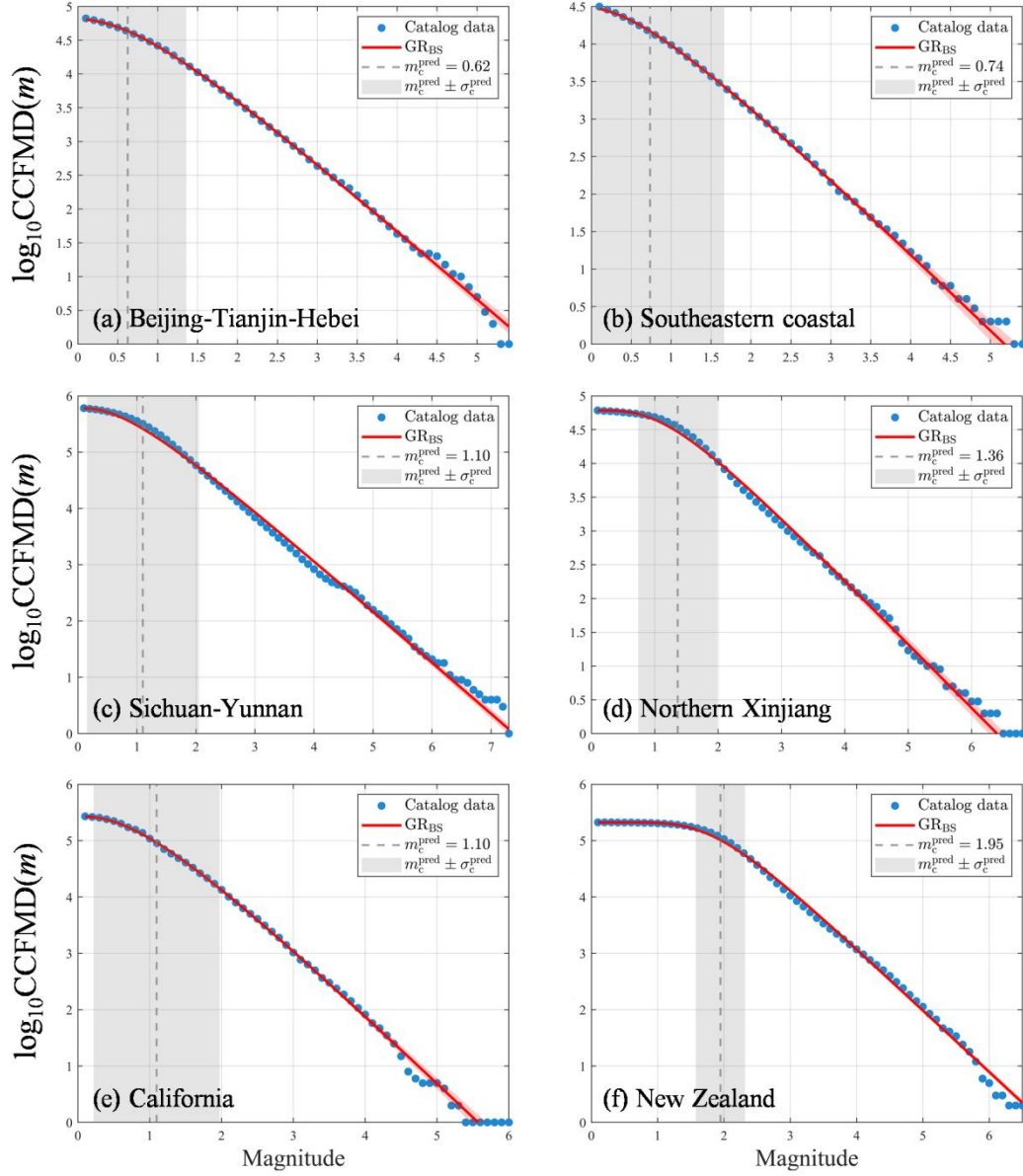


Figure S7. CCFMD of the empirical earthquake catalogs from six study regions, along with the fitted CCFMD(m) derived using the augmented Gutenberg-Richter law with the BSR_{LU}(x) function (GR_{BS}). The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

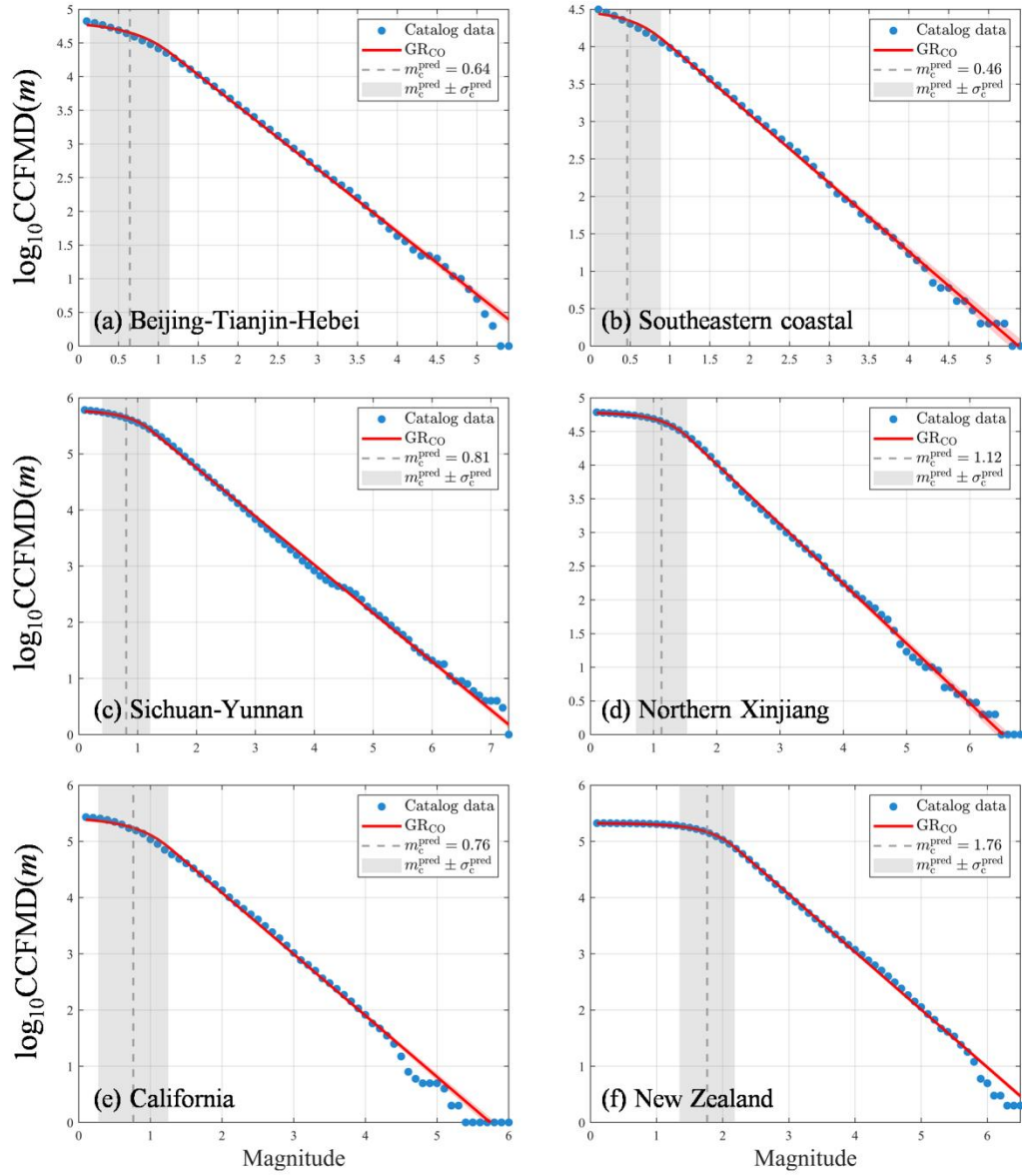


Figure S8. CCFMD of the empirical earthquake catalogs from six study regions, along with the fitted CCFMD(m) derived using the augmented Gutenberg-Richter law with the CORELU(x) function (GR_{CO}). The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

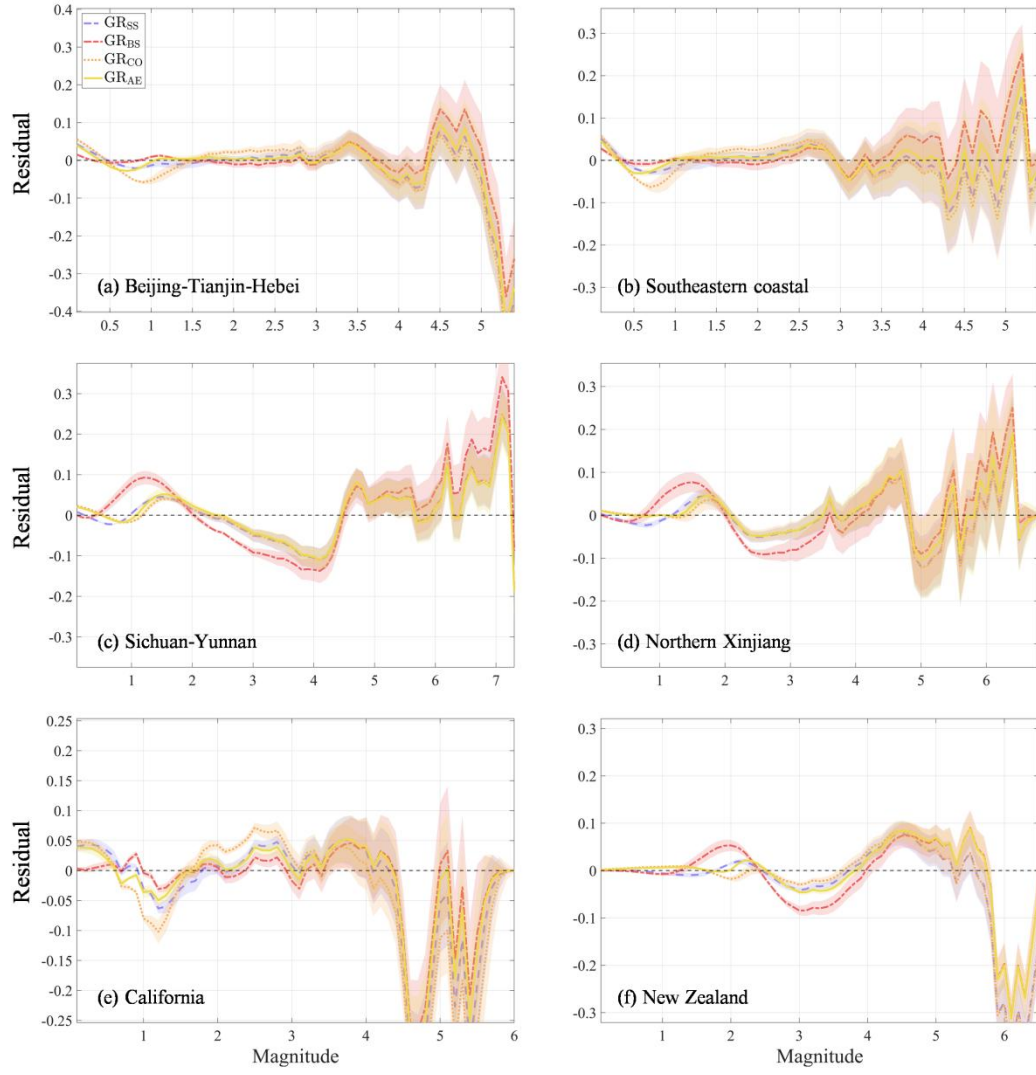


Figure S9. Residuals between the empirical $\text{CCFMD}^{\text{obs}}(m)$ and the fitted augmented Gutenberg-Richter models with four $\text{gReLU}(x)$ functions, $\text{CCFMD}^{\text{fit}}(m)$, expressed as $\log_{10} \text{CCFMD}^{\text{obs}}(m) - \log_{10} \text{CCFMD}^{\text{fit}}(m)$, for the six study regions. Shaded areas indicate the uncertainty of the residuals, estimated from 200 bootstrap iterations.